Visit **www.brpaper.com** for downloading previous years question papers of
10th and 12th (PSEB and CBSE), IKPTU,MRSSTU, PSBTE, PANJAB UNIVERSITY,
PUNJABI UNIVERSITY, BFUHS, HPTU, HPSBTE , HARYANA DIPLOMA, MDU HARYANA

**Roll No.** | | | | | | | | | | | | | **Total No. of Pages : 02**

**Total No. of Questions : 18**

**B.Tech. (Electrical & Electronics) (OE 2013 Onwards Batch)**
**(Sem.–6)**

# TRANSDUCERS AND SIGNAL CONDITIONING

**Subject Code : BTEEE-OPD**
**M.Code : 72841**

**Time : 3 Hrs.** **Max. Marks : 60**

**INSTRUCTIONS TO CANDIDATES :**

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains FIVE questions carrying FIVE marks each and students have to attempt any FOUR questions.**
3. **SECTION-C contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.**

## SECTION-A

**Answer briefly :**

1. What is the basic operating principle of digital tape recording?

2. Give the principle of capacitive transducers.

3. What is thermistor? State the advantage and disadvantages.

4. What are the advantages of LCD over LED?

5. Write down the potentials for pre-accelerating and accelerating electrodes?

6. Define transducer and inverse transducer.

7. Draw the diagram of integrator and differentiator using OP-Amp.

8. List the types of ADC and DAC?

9. What are the two physical parameters in strain gauge?

10. Differentiate between modulator and demodulator.

brpaper.com
brpaper.com



BR paper
exams preparatory

## SECTION-B

11. What are the selection criteria for the transducer? Explain the working principle of LVDT with neat sketch and characteristics. Give advantages, disadvantages and applications of LVDT.

12. Classify the different types of telemetering systems? Explain the landline telemetering system and describe its advantages and disadvantages.

13. Derive the expression for closed loop gain of an operational amplifier used in non-inverting mode. Describe the assumption made.

14. What is an XY recorder? How do you distinguish it from Xt and Yt recorders?

15. Draw and explain current to voltage Converter with neat diagram.

## SECTION-C

16. Describe the basic principle of strain gauge, its types, advantages and disadvantages. Also explain how strain gauge is used for pressure and torque measurements?

17. a) Describe the working of Time division multiplexing with neat sketch.

    b) Explain the internal structure of CRT and describe the principle of electrostatic focusing.

18. Write short notes on following :

    a) Analog modulator and demodulator

    b) Capacitive Transducers

brpaper.com
brpaper.com